CDAC Admission Process Case Study :-

# Design database for CDAC Admission System.

# Create appropriate tables and insert sample records.

# Write a Stored Procedure for Round1 allocation.

# Write a Stored Procedure for Round2 allocation.

# Note:

1. Create ER diagram following Normalization process.

2. Study database constraints in depth: PRIMARY KEY, FOREIGN-KEY, UNIQUE, CHECK, DEFAULT Apply appropriate constraints in CDAC admission system.

3. Learn about composite keys and candidate keys Identify keys in CDAC admission system.

4. Explore AUTO_INCREMENT / SEQUENCES Use appropriate AUTO_INCREMENT keys in CDAC admission system.

5. Study control structures: IF-ELSE, CASE, LOOPS (WHILE, REPEAT, LOOP) Use appropriate control structures in CDAC admission system stored procedures.

6. Learn stored procedure parameters: IN, OUT, IN-OUT. Use appropriate parameters for CDAC admission system stored procedures.

7. Understand cursors and their usage in following stored procedures. Make appropriate use of cursor in CDAC admission system stored procedures. 8. Explore error handling in stored procedures Implement appropriate error handling in CDAC admission system stored procedures.

Creating tables and inserting records :-

```
d2_96712_Devanand> CREATE TABLE ccat_result (
    ->     form_no INT PRIMARY KEY,
    ->     name VARCHAR(100) NOT NULL,
    ->     AIRank INT UNIQUE,
    ->     total_marks INT CHECK (total_marks >= 0),
    ->     created_at TIMESTAMP DEFAULT NOW()
    -> );
Query OK, 0 rows affected (1.360 sec)

d2_96712_Devanand> CREATE TABLE centers (
    ->     center_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     center_name VARCHAR(100) NOT NULL UNIQUE,
    ->     address VARCHAR(200)
    -> );
Query OK, 0 rows affected (0.478 sec)

d2_96712_Devanand> CREATE TABLE courses (
    ->     course_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     course_name VARCHAR(100) NOT NULL UNIQUE,
    ->     fees DECIMAL(10,2) CHECK (fees > 0)
    -> );
Query OK, 0 rows affected (0.697 sec)

d2_96712_Devanand> CREATE TABLE center_courses (
    ->     offer_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     center_id INT,
    ->     course_id INT,
    ->     capacity INT CHECK (capacity >= 0),
    ->
    ->     FOREIGN KEY (center_id) REFERENCES centers(center_id),
    ->     FOREIGN KEY (course_id) REFERENCES courses(course_id),
    ->
    ->     UNIQUE(center_id, course_id) -- composite candidate key
    -> );
Query OK, 0 rows affected (0.596 sec)
```

```
d2_96712_Devanand> CREATE TABLE preferences (
    ->     pref_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     form_no INT,
    ->     offer_id INT,
    ->     pref_no INT CHECK (pref_no IN (1,2)),
    ->     freeze BOOLEAN DEFAULT FALSE,
    ->
    ->     FOREIGN KEY (form_no) REFERENCES ccat_result(form_no),
    ->     FOREIGN KEY (offer_id) REFERENCES center_courses(offer_id),
    ->
    ->     UNIQUE(form_no, pref_no) -- composite key
    -> );
Query OK, 0 rows affected (0.788 sec)

d2_96712_Devanand> CREATE TABLE allocations (
    ->     alloc_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     form_no INT,
    ->     offer_id INT,
    ->     round_no INT,
    ->     status ENUM('ALLOCATED','UPGRADED','RETAINED'),
    ->
    ->     FOREIGN KEY (form_no) REFERENCES ccat_result(form_no),
    ->     FOREIGN KEY (offer_id) REFERENCES center_courses(offer_id)
    -> );
Query OK, 0 rows affected (0.745 sec)

d2_96712_Devanand> INSERT INTO ccat_result VALUES
    -> (101,'Amit',10,85,NOW()),
    -> (102,'Ravi',5,90,NOW()),
    -> (103,'Neha',20,75,NOW()),
    -> (104,'Priya',8,88,NOW()),
    -> (105,'Karan',15,80,NOW()),
    -> (106,'Anjali',12,82,NOW());
Query OK, 6 rows affected (0.212 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
d2_96712_Devanand> INSERT INTO centers (center_name,address) VALUES
    -> ('Pune','Pune , Maharashtra'),('Mumbai','Mumbai Maharashtra'),('NOida','noida uttar pradesh');
Query OK, 3 rows affected (0.183 sec)
Records: 3  Duplicates: 0  Warnings: 0

d2_96712_Devanand> INSERT INTO courses (course_name,fees) VALUES
    -> ('DAC',80000),('DBDA',90000),('DESD',85000);
Query OK, 3 rows affected (0.171 sec)
Records: 3  Duplicates: 0  Warnings: 0

d2_96712_Devanand> INSERT INTO center_courses(center_id,course_id,capacity) VALUES
    -> (1,1,2),(1,2,2),(2,1,2),(3,2,2);
Query OK, 4 rows affected (0.182 sec)
Records: 4  Duplicates: 0  Warnings: 0

d2_96712_Devanand> INSERT INTO ccat_result VALUES
    -> (1,'A',1,90,NOW()),
    -> (2,'B',2,85,NOW()),
    -> (3,'C',3,80,NOW()),
    -> (4,'D',4,75,NOW());
Query OK, 4 rows affected (0.145 sec)
Records: 4  Duplicates: 0  Warnings: 0

d2_96712_Devanand> INSERT INTO preferences(form_no,offer_id,pref_no) VALUES
    -> (101,1,1),
    -> (102,1,1),(102,3,2),
    -> (103,2,1),
    -> (104,3,1),
    -> (104,1,2),
    -> (105,1,1),
    -> (105,2,2),
    -> (106,3,1);
Query OK, 9 rows affected (0.168 sec)
Records: 9  Duplicates: 0  Warnings: 0
```

Round 1st allocations :-

```
d2_96712_Devanand> CREATE PROCEDURE round1_alloc()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->
    ->     DECLARE v_form INT;
    ->     DECLARE v_offer INT;
    ->     DECLARE v_capacity INT;
    ->     DECLARE cur CURSOR FOR SELECT form_no FROM ccat_result ORDER BY AIRank;
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->     OPEN cur;
    ->     read_loop: LOOP
    ->         FETCH cur INTO v_form;
    ->         IF done THEN
    ->             LEAVE read_loop;
    ->         END IF;
    ->
    ->         SELECT offer_id INTO v_offer
    ->         FROM preferences
    ->         WHERE form_no = v_form AND pref_no = 1;
    ->
    ->         SELECT capacity INTO v_capacity
    ->         FROM center_courses
    ->         WHERE offer_id = v_offer;
    ->
    ->         IF v_capacity > 0 THEN
    ->             INSERT INTO allocations(form_no,offer_id,round_no,status)
    ->             VALUES(v_form,v_offer,1,'ALLOCATED');
    ->
    ->             UPDATE center_courses
    ->             SET capacity = capacity - 1
    ->             WHERE offer_id = v_offer;
    ->
    ->         ELSE
    ->             SELECT offer_id INTO v_offer
    ->             FROM preferences
    ->             WHERE form_no = v_form AND pref_no = 2;
    ->
    ->             IF v_offer IS NOT NULL THEN
    ->                 SELECT capacity INTO v_capacity
    ->                 FROM center_courses
    ->                 WHERE offer_id = v_offer;
    ->
    ->                 IF v_capacity > 0 THEN
    ->                     INSERT INTO allocations(form_no,offer_id,round_no,status)
    ->                     VALUES(v_form,v_offer,1,'ALLOCATED');
    ->
    ->                     UPDATE center_courses
    ->                     SET capacity = capacity - 1
    ->                     WHERE offer_id = v_offer;
    ->                 END IF;
    ->             END IF;
    ->         END IF;
    ->     END LOOP;
    ->     CLOSE cur;
    -> END;
    -> $$
Query OK, 0 rows affected (0.333 sec)
```

Round 2 nd  allocations :-

```
d2_96712_Devanand> DELIMITER ;
d2_96712_Devanand> DELIMITER $$
d2_96712_Devanand>
d2_96712_Devanand> CREATE PROCEDURE round2_alloc()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE v_form INT;
    ->     DECLARE v_curr_offer INT;
    ->     DECLARE v_new_offer INT;
    ->     DECLARE v_capacity INT;
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT a.form_no, a.offer_id
    ->         FROM allocations a
    ->         JOIN preferences p ON a.form_no = p.form_no
    ->         WHERE a.round_no = 1 AND p.freeze = FALSE;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     read_loop: LOOP
    ->         FETCH cur INTO v_form, v_curr_offer;
    ->         IF done THEN
    ->             LEAVE read_loop;
    ->         END IF;
    ->
    ->         -- try higher preference (pref 1)
    ->         SELECT offer_id INTO v_new_offer
    ->         FROM preferences
    ->         WHERE form_no = v_form AND pref_no = 1;
    ->
    ->         IF v_new_offer != v_curr_offer THEN
    ->
    ->             SELECT capacity INTO v_capacity
    ->             FROM center_courses
    ->             WHERE offer_id = v_new_offer;
    ->
    ->             IF v_capacity > 0 THEN
    ->                 -- upgrade
    ->                 INSERT INTO allocations(form_no,offer_id,round_no,status)
    ->                 VALUES(v_form,v_new_offer,2,'UPGRADED');
    ->
    ->                 UPDATE center_courses
    ->                 SET capacity = capacity - 1
    ->                 WHERE offer_id = v_new_offer;
    ->
    ->                 UPDATE center_courses
    ->                 SET capacity = capacity + 1
    ->                 WHERE offer_id = v_curr_offer;
    ->             ELSE
    ->                 -- retain
    ->                 INSERT INTO allocations(form_no,offer_id,round_no,status)
    ->                 VALUES(v_form,v_curr_offer,2,'RETAINED');
    ->             END IF;
    ->
    ->         END IF;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    -> END$$
Query OK, 0 rows affected (0.167 sec)
```